

Blood Work Analysis — Roman

Draw of June 8, 2026 | Report generated June 9, 2026
Sources: yesterday's draw, 10-yr consolidated history, active 31-compound protocol & 30-day adherence, Function Health panel, and parallel literature review (PubMed + ClinicalTrials.gov)

■ **Not medical advice.** This is a data + literature synthesis to bring to your physician. Several panels (iron, testosterone) were still “in progress” at report time, so conclusions are preliminary.

TL;DR. A reassuring panel with a genuinely excellent trend story. The four flags are either **longstanding / constitutional** (macrocytosis) or **clinically trivial** (trace leukocyte esterase). Nothing acute. The headline is in the *trends*, not the flags — and much of it is plausibly attributable to your protocol, in the direction you'd want.

1. What flagged yesterday (Jun 8)

Marker	Value	Range	Flag	Verdict
RBC	4.02	4.20–5.80	LOW	Longstanding mild — see §3A
MCV	102.5	81.4–101.7	HIGH	Constitutional, stable since 2016
MCH	33.6	27.0–33.0	HIGH	Rides along with MCV
Leukocyte esterase (urine)	TRACE	Negative	flag	Clinically minor — see §3D

Everything else was clean: WBC 5.4, platelets 250, full differential normal, Hgb 13.5, Hct 41.2, RDW 14.4; urine protein / blood / glucose / nitrite all negative; **ferritin 62** (normal). Iron/TIBC/%sat and the testosterone panel were still **in progress** — get those before drawing firm conclusions.

2. The trend story (the headline)

Your cardiometabolic profile has transformed over ~6 months:

Marker	Dec 2025	Latest 2026	Δ
Total cholesterol	220	141	–79
LDL	135	77	–58
ApoB	97	69	–28
Triglycerides	128	37	–91
Non-HDL	148	89	–59
hs-CRP	0.6	0.2	very low inflammation
Insulin (uIU/mL)	2.5	2.8	excellent sensitivity
HbA1c (%)	5.4	5.3	optimal
Testosterone total (ng/dL)	261 (Sep'25)	616 (Apr'26)	optimized off low baseline
Free T (pg/mL)	33.1	48.6	now in range

Liver enzymes **were** stable here (AST 44, ALT 29) — at the time read as muscle/training source. ■ **This was superseded on Jun 8 by a hepatocellular liver-injury pattern — see the Jun 9 Addendum (final page).** Kidney function stable (eGFR ~79–95). Two minor trade-offs: HDL drifted down (76 → 52) and Lp(a) read lower (88 → 58) — both explained below; neither is a concern.

3. Can it be attributed to peptides? — the core question

Each finding was tested against the literature. Bottom line: the favorable trends are well-explained by your stack, and the “concerning” CBC findings are explained by it too — in a protective way.

A) The low-RBC-despite-TRT paradox → most likely GH-axis hemodilution

On ~140 mg/wk testosterone you'd expect high hematocrit. Instead RBC is mildly low and Hct normal. Two opposing forces:

- **GH secretagogues (Tesamorelin + Ipamorelin) and IGF-1 LR3 expand plasma/extracellular fluid volume** (sodium/water retention), diluting concentration-based indices — a documented GH-class effect.
- **Daily SubQ micro-dosed testosterone is the least erythrocytosis-prone route** — steady, no supraphysiologic peaks. Peak-driven IM ~triples erythrocytosis risk.

Møller, JCEM 1991 (PMID 1826008); tesamorelin RCTs Falutz PMID 20101189, Stanley JAMA 2014; Madsen, JCEM 2021 (PMID 33599731); Wilson, AJHP 2018 (PMID 29367424).

Net: the GH/IGF axis is masking the erythrocytosis TRT would otherwise cause — protective (lower thrombosis risk). With normal-for-you MCV, replete iron, and a stable 9-month trend, this is **not a true anemia**. **Flip side:** if GH agents are tapered, watch for Hct rising into erythrocytotic range. **Confirm with a reticulocyte count** ($\pm$ LDH/haptoglobin).

B) Macrocytosis (MCV/MCH) → constitutional, NOT peptide-driven

MCV has been ~98–102 on **every draw since 2016** — it predates the entire stack. B12 normal (485–642), no anemia, only mildly elevated (vs the 109–113 typical of megaloblastic/MDS). Classic **constitutional macrocytosis**. No human evidence links your NAD-pathway compounds (NAD⁺, 5-Amino-1MQ, 6-methoxynicotinamide, ATX-304, SS-31, MOTS-c) to raised MCV, and the timing (macrocytosis first) argues against causation. **Confirm benign** with reticulocyte, RDW (already 14.4, reassuring), peripheral smear, MMA + homocysteine, RBC folate, TSH, and an alcohol-intake check. If normal → no marrow biopsy, nothing to do. (*Colon-Otero, Med Clin North Am 1992, PMID 1578958.*)

C) The lipid transformation → yes, attributable to Retatrutide + weight loss

Magnitude matches the **Retatrutide phase 2 trial** (NCT04881760; Jastreboff, NEJM 2023), which produced 17–24% weight loss with tightly-tracking lipid improvements. The –91 triglyceride drop is the signature of glucagon/GIP triple-agonism plus your energy-deficit state (leptin 0.4, insulin 2.5). ApoB → 69 confirms a genuine atherogenic-particle reduction, not a calc artifact. Cagrilintide + caloric restriction additive. Research compounds (SLU-PP-332, MOTS-c, 5-Amino-1MQ) have **no human lipid trials** — don't credit them.

- **HDL 76→52:** expected, benign — reciprocal of a collapsed-TG state during active rapid weight loss; usually settles at maintenance. With ApoB 69 / TG 37, risk profile is *better*, not worse.
- **Lp(a) 88→58: do not attribute to peptides** — Lp(a) is ~90% genetic and isn't lowered by incretins. Assay/lab noise across portals. Re-measure once on a single consistent assay.

D) Leukocyte esterase TRACE (urine) → clinically minor

Trace LE with **negative nitrite, protein, and blood, clear urine, asymptomatic** is most likely contamination or incidental low-grade pyuria. IDSA 2019 (Nicolle, PMID 30895288): don't screen/treat asymptomatic bacteriuria in men — **no antibiotics indicated**. If symptom-free, just repeat a clean-catch midstream sample at routine follow-up; pursue micro + culture only if persistent or symptomatic.

4. Suggested next steps (to bring to your physician)

Already pending — just get results: iron panel (TIBC/%sat) and testosterone/SHBG/free-T.

Cheap confirmatory adds that close every open question:

- **Reticulocyte count** — resolves the low-RBC question (diluted vs true anemia).
- **MMA + homocysteine, RBC folate, TSH** — confirms macrocytosis is benign/constitutional.
- **Single-assay Lp(a) recheck** — establish true genetic baseline; stop tracking cross-portal noise.
- **Repeat clean-catch UA** if you want to clear the trace LE (optional if asymptomatic).

Monitoring notes:

- Keep periodic CBCs — if GH agents are reduced, Hct could rise (TRT erythrocytosis surveillance).
- AST 44 is likely muscle — confirm by drawing after ≥ 48 h rest from heavy training (consider CK/GGT).
- You're in an aggressive energy-deficit phase (leptin 0.4, falling HDL, very low TG) — protect lean mass with protein + resistance training through the retatrutide taper.

Honest summary. Nothing in yesterday's results is alarming. Two flags are constitutional/longstanding, one is trivial, and the trend lines — lipids, glucose/insulin, inflammation, testosterone — are genuinely impressive and consistent with what your protocol is designed to do. The most “abnormal” CBC finding (low RBC despite TRT) is actually a signature of your GH/IGF + daily-SubQ-T design keeping you out of erythrocytosis.

ADDENDUM — Acute Liver-Enzyme Update

Added June 9, 2026, after a separate liver panel from the Jun 8 draw. This section supersedes the “AST = muscle” note in §2.

What changed. A liver panel from the same Jun 8 draw shows a **hepatocellular injury pattern** that was not in the earlier images. This is the one finding that warrants prompt action.

Marker	Jun 8	Ref	Read
ALT	148	≤55	HIGH — flagged urgent (~2.7× ULN)
AST	103	≤48	HIGH (~2× ULN)
ALP	38	low-normal	Argues against cholestasis
Total bilirubin	0.5	0.2–1.2	Normal — see kinetics note

Context: on **May 5** your AST/ALT were 44/29 (near-normal). The rise happened **May 5 → Jun 8**, coinciding with new agents — not over the prior 3 months.

Most likely cause (hepatocellular DILI)

- **LGD-4033 (SARM) — prime suspect.** Actual dosing **10 mg/day × ~15–17 days** (≈10× the highest dose ever tested for human safety, Basaria 2013), timing textbook for SARM drug-induced liver injury. **Stopped.**
- **NNMT-inhibitor orals (5-Amino-1MQ, JBSNF-0008) — plausible co-contributors;** they act directly on hepatic methylation/NAD pathways. **Stop.**
- **Sertraline — low probability.** Recognized but uncommon DILI cause; your May 5 LFTs were normal while already ~2 months in, which largely exonerates it. **Already stopped ~Jun 2.**
- Other research orals (J-147, ATX-304, SLU-PP-332): no human data — stop. Finasteride: low suspicion.

How worried to be — calibrated

Normal bilirubin means you **caught this early**, not that you're guaranteed safe — bilirubin is a *lagging* indicator that rises only after substantial functional loss. Treat this as an **actively evolving** injury. Because of tissue-saturation kinetics, transaminases can keep **rising for 1–3+ weeks** after stopping the drug; a higher ALT next week is expected lag, **not** automatic failure. Danger is judged by **INR and bilirubin**, not the transaminase number. Two of the top suspects (LGD + sertraline) are already removed, so the dechallenge is underway.

Symptom matrix — what to monitor over the next ~14 days

Your symptom gauge is unreliable right now because SSRI-discontinuation symptoms overlap liver symptoms. Separate them rigidly:

Bucket	Symptoms	Action
■ SSRI withdrawal	"Brain zaps", vertigo/dizziness, sensory changes, emotional lability/tearfulness	Manage with prescriber. Do NOT self-restart sertraline
■ Confounded — useless for liver	Nausea, fatigue, lethargy, muscle aches, sweating, diarrhea	Expect these (withdrawal + early hepatitis). Do NOT use to judge the liver — rely on labs
■ Liver-specific — ER NOW	Scleral icterus (check eye-whites daily), pale/clay stools, dark tea/cola urine, itching without rash, RUQ pain, easy bruising/bleeding gums, confusion	Immediate ER

Tests — within ~24–48 h (don't wait a week)

- **PT/INR ■** — the real safety/synthetic-function marker — plus **total bilirubin** and **albumin**.
- **GGT + CK** — confirm liver-vs-muscle source.
- **ALT/AST/ALP** repeat — expect possibly higher; judge danger by INR/bilirubin, not the number.
- **Viral hepatitis A/B/C (±E)** — exclude a coincidental cause.
- An **at-home kit (e.g. Rythm)** is **NOT sufficient** for this — it can't do PT/INR, GGT, CK, or hepatitis serologies, and turnaround is too slow. Use a conventional venous draw; the kit is fine only as a parallel longevity re-check.

Additional watch-items flagged in review

- **Hypoglycemia awareness.** **IGF-1 LR3** is insulin-mimetic and—with your aggressive energy deficit—can drive lows independent of the liver (the “failing-liver gluconeogenesis” mechanism is not yet in play at this mild stage). Keep a **fingerstick glucometer** as backup; don't rely on a consumer CGM to catch a severe low. Discuss **pausing**

non-essential metabolic injectables (esp. IGF-1 LR3) with your physician while the liver recovers.

- **Hormone/mood shift.** LGD-4033 strongly suppresses SHBG (Basaria 2013); with exogenous testosterone, total T stays up but **free testosterone/estradiol can swing** as SHBG recovers over 2–4 weeks — possible mood volatility that may overlap SSRI withdrawal. Anticipate it; not dangerous.
- **SSRI symptom relief.** If withdrawal becomes intolerable, **gabapentinoids (gabapentin/pregabalin)** are renally cleared (no hepatic metabolism) and are a physician-managed, liver-safe option to ask about — rather than reinstating sertraline.

Bottom line. Active, early hepatocellular drug-induced liver injury — LGD-4033 (10 mg/day) the clear trigger. Keep all suspect orals stopped, get **PT/INR + hepatic panel + GGT/CK + hepatitis** within 24–48 h, monitor only the **liver-specific** red flags, don't self-restart sertraline, and loop in your physician promptly (liver + SSRI taper + whether to pause metabolic injectables).

Generated for Roman Semenov, June 9, 2026. Literature synthesis via PubMed and ClinicalTrials.gov. This document is informational and does not constitute medical advice or a diagnosis; review all findings and any testing decisions with your treating clinician.